

Practice Problem Set 1

1. A biased random walk process is defined as $X_n = \sum_{i=0}^n U_i$, $n \geq 0$
 where $\{U_i, i \geq 0\}$ is an i.i.d Bernoulli process with PMF of U_i being $p_{U_i}(k) = \begin{cases} \frac{1}{4} & k=-1 \\ \frac{3}{4} & k=1 \\ 0 & \text{else} \end{cases}$
 Find $\mu_X[n]$ and $\text{Var}_X[n]$
2. A random process $\{X_n: n \in \mathbb{N}\}$ is known to be W.S.S with $\mu_X[0] = 10$ and $\text{var}_X[0] = 1$. Determine $\mu_X[100]$ and $\text{var}_X[100]$.
3. Consider an i.i.d Bernoulli process $\{X_n: n \in \mathbb{N}\}$ where $X_i \sim \text{unif}\{0, 1\}$
 Let $Y_n = (-1)^n X_n$, $n \in \mathbb{N}$. Is $\{Y_n: n \in \mathbb{N}\}$ an i.i.d process? Justify.
4. Consider the continuous parameter process $\{X_t: t \in \mathbb{R}\}$, $X_t = A \cos 2\pi t$, $t \in \mathbb{R}$
 where A is a random variable $\sim N(0, 1)$. Find $\mu_X(t)$ and $\text{Cor}_X(s, t)$.
5. Consider a random process $\{X_n: n \in \mathbb{Z}\}$ defined as $X_n = a_0 U_n + a_1 U_{n-1}$,
 where a_0, a_1 are constants and $\{U_n: n \in \mathbb{Z}\}$ is an i.i.d process with mean $\mu_U[n] = 0 \forall n \in \mathbb{Z}$ and $\text{var}_U[n] = 1 \forall n \in \mathbb{Z}$. Is $\{X_n: n \in \mathbb{Z}\}$ WSS?
 If yes, find $\mu_X[n]$ and $R_X[\Delta]$.
6. A random process $\{X_n: n \in \mathbb{Z}\}$ consists of independent random variables
 with $X_n \sim \begin{cases} N(0, 1) & \text{for } n \text{ even} \\ \text{unif}(-\sqrt{3}, \sqrt{3}) & \text{for } n \text{ odd} \end{cases}$.
 Is this process WSS? Is it stationary?
7. The random processes $\{X_n: n \in \mathbb{N}\}$ and $\{Y_n: n \in \mathbb{N}\}$ are both WSS. Further $X_n \perp Y_m \forall n, m \in \mathbb{N}$.
 a) Is $\{Z_n = X_n + Y_n: n \in \mathbb{N}\}$ WSS? b) Is $\{W_n = X_n Y_n: n \in \mathbb{N}\}$ WSS?